



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to LAKSHMI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

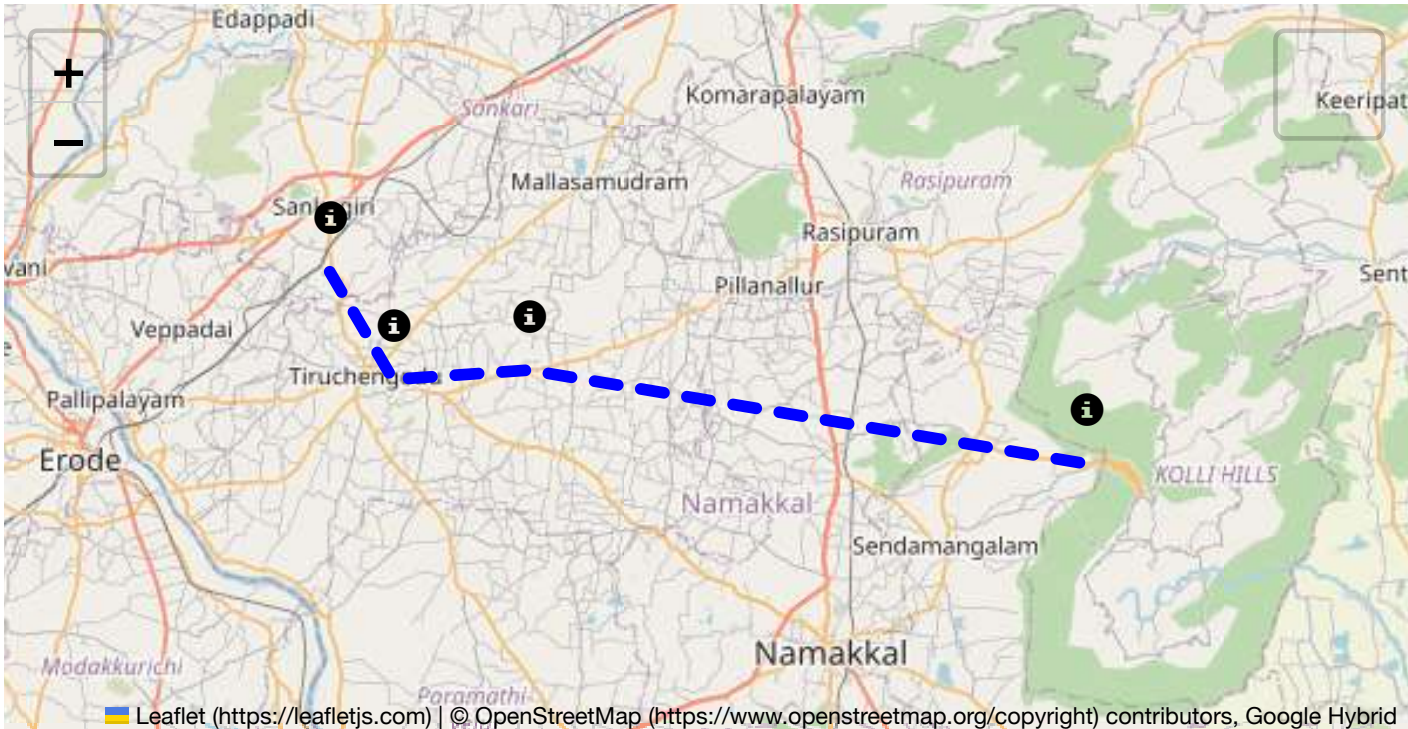
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

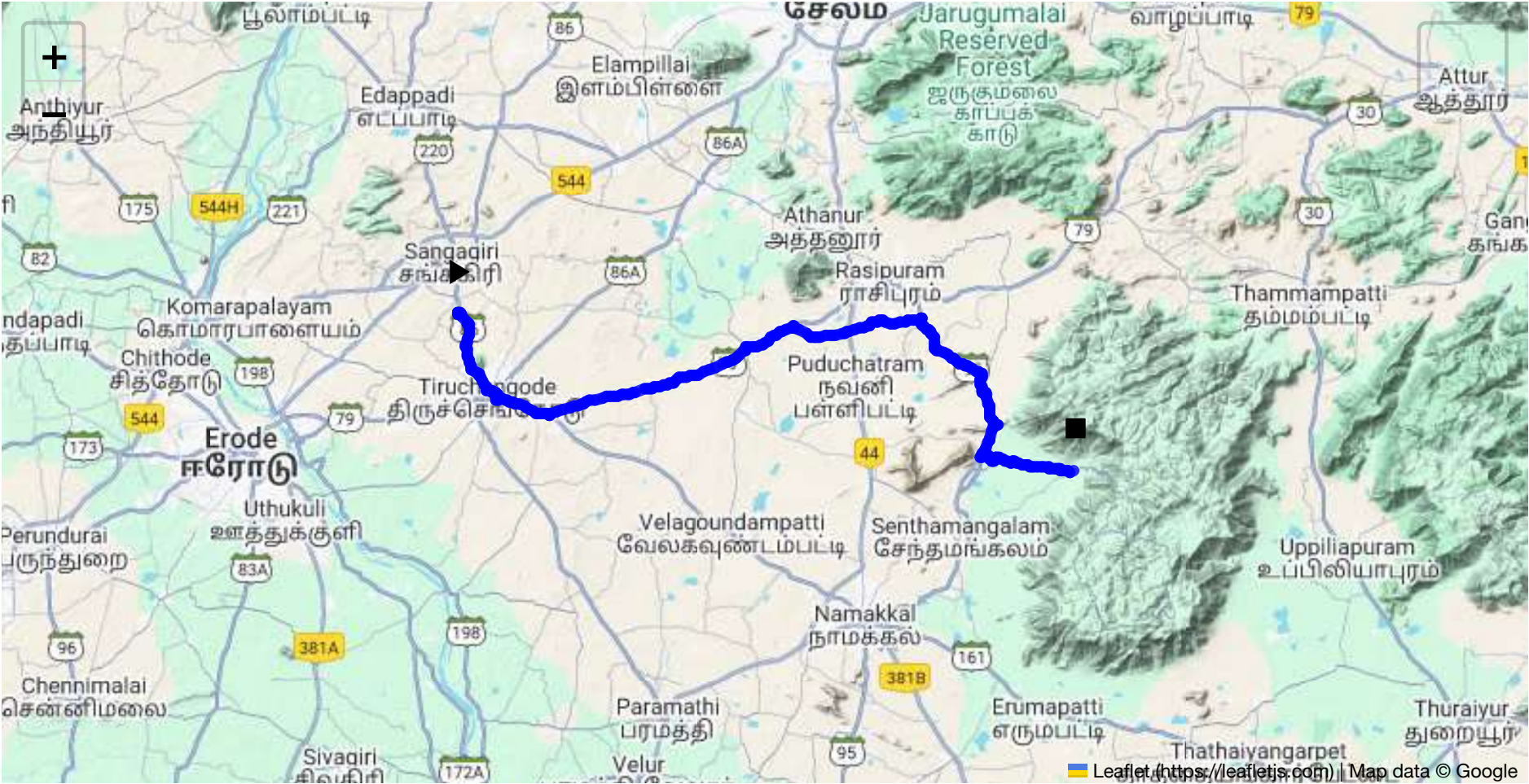
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 67.41 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.9 hours

Start: (11.4381, 77.8734)
End: (11.327316, 78.317764)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Karavalli via specified waypoints intersects several local roads and a state highway (SH 79), covering around 67.41 kilometers. The journey navigates through both rural and semi-urban areas in Tamil Nadu. The trip takes approximately 1.52 hours for heavy vehicles, suggesting stops and slower travel due to terrain and traffic conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate, with hot and dry conditions, especially from March to June. The southwest and northeast monsoons from June to September can lead to heavy rainfall. Hazards may include reduced visibility, slippery roads, and possible flooding in low-lying areas during monsoons.

3. Analysis of Traffic Patterns

Traffic tends to be heavier during early morning (7-9 AM) and evening (5-7 PM) hours, especially near towns and markets. Congestion-prone areas include town centers and intersections with SH 79, where local traffic and heavy vehicles merge.

4. Assessment of Road Quality and Infrastructure

The main highway (SH 79) is generally well-maintained, with clear signage and marked lanes. However, smaller connecting roads may have varying conditions, including potholes, uneven surfaces, and narrower lanes that can pose challenges for heavy vehicles, especially during rains.

5. Suggestions for Alternative Routes for Emergencies

- An alternate longer route could bypass some smaller roads by taking an extended path via major highways, possibly reducing some risks but increasing travel time.
- Local knowledge should be sought for real-time detours in case of roadblocks or closures.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is regulated by the Tamil Nadu Motor Vehicle Rules, aligning with Central Motor Vehicle Rules. Permits and vehicle documentation must be valid, and transport at night may be restricted, depending on the material type and risk level.

7. Overview of Historical Incidents

There aren't frequent significant recorded incidents specific to this route, but occasional accidents involving heavy vehicles are reported in the region due to hilly areas and sharp bends, requiring cautious driving.

8. Environmental Considerations and Sensitive Areas

Passing through semi-urban regions, precautions are required near water bodies and agricultural zones to prevent contamination from spills. Waste disposal and refueling must be managed to minimize environmental impact.

9. Analysis of Communication Coverage

Urban and semi-urban segments along the route generally have good mobile network coverage. There may be patches of weak signal, particularly in more rural areas and valleys, impacting communication negatively.

10. Estimated Emergency Response Times

Response times may vary, averaging 30-45 minutes in towns due to local emergency services. Rural areas may experience longer times due to distance from service centers.

11. Overall Summary of Risk Assessment

Overall, the route is moderately challenging, mainly due to variable road conditions and unpredictable weather. Risks revolve around poor road quality in sections, congestion, and environmental sensitivities. Adequate preparation, adherence to regulation, and responsive planning can mitigate most identified risks efficiently. Regular updates and coordination with local authorities are recommended for effective hazard management.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38295, 77.90001	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	Medium	11.36745, 77.93951	30 KM/Hr
10	Turn	Medium	11.41321, 78.08067	30 KM/Hr
11	Turn	High	11.41453, 78.08062	15 KM/Hr
12	Turn	Medium	11.41476, 78.08094	30 KM/Hr
13	Turn	High	11.42962, 78.11578	15 KM/Hr
14	Turn	Medium	11.42943, 78.11608	30 KM/Hr
15	Turn	High	11.42938, 78.11609	15 KM/Hr
16	Turn	Medium	11.42450, 78.15278	30 KM/Hr
17	Turn	Medium	11.43142, 78.18808	30 KM/Hr
18	Blind Spot	Blind Spot	11.43542, 78.20754	10 KM/Hr
19	Turn	Medium	11.43050, 78.20854	30 KM/Hr
20	Turn	Medium	11.42751, 78.21298	30 KM/Hr
21	Turn	High	11.42178, 78.21699	15 KM/Hr
22	Turn	High	11.39529, 78.25156	15 KM/Hr
23	Turn	Medium	11.38226, 78.25407	30 KM/Hr
24	Turn	Medium	11.36047, 78.25982	30 KM/Hr
25	Turn	Medium	11.35997, 78.26190	30 KM/Hr
26	Blind Spot	Blind Spot	11.35989, 78.26201	10 KM/Hr
27	Turn	Medium	11.35862, 78.25971	30 KM/Hr
28	Blind Spot	Blind Spot	11.33761, 78.25054	10 KM/Hr
29	Turn	Medium	11.33765, 78.25083	30 KM/Hr
30	Turn	Medium	11.33897, 78.25450	30 KM/Hr
31	Turn	Medium	11.33831, 78.25543	30 KM/Hr
32	Turn	Medium	11.33675, 78.25633	30 KM/Hr
33	Turn	High	11.33596, 78.25642	15 KM/Hr
34	Turn	High	11.33556, 78.25816	15 KM/Hr
35	Blind Spot	Blind Spot	11.33448, 78.25806	10 KM/Hr
36	Turn	Medium	11.33666, 78.26098	30 KM/Hr
37	Turn	Medium	11.33697, 78.26238	30 KM/Hr
38	Turn	Medium	11.33371, 78.27258	30 KM/Hr
39	Turn	Medium	11.32983, 78.29999	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
40	Turn	High	11.32944, 78.30032	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38342, 77.89975



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38295, 77.90001



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36745, 77.93951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41321, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41476, 78.08094



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42962, 78.11578



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42943, 78.11608



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42938, 78.11609



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42450, 78.15278



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43142, 78.18808



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43050, 78.20854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42751, 78.21298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42178, 78.21699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39529, 78.25156



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38226, 78.25407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36047, 78.25982



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.35997, 78.26190



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.35989, 78.26201



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.35862, 78.25971



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.33761, 78.25054



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33765, 78.25083



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.33448, 78.25806



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.32983, 78.29999



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.32944, 78.30032
